

Xiangyuan(Shawn) Mo

Mobile: (+1)919-2014-212; E-mail: xm89@duke.edu; GitHub: <https://github.com/RUC-Tree>

EDUCATION

- Master of Arts In Economics, Duke University, NC, United States** 08/2025 - 05/2027
- **Research Interests:** Econometrics, Casual Inference; **GPA:** 4.00/4.00
 - **Core Courses:** Real Analysis (A+), Econometrics (A+), Predictive Modeling and Statistical Learning (A+)
 - Bachelor of Economics, Renmin University of China (RUC), Beijing, China** 09/2021 - 06/2025
 - **Major:** Quantitative Economics; **GPA:** 3.80/4.00
 - **Awards:** Best model in the first RUC Quantitative Modeling Competition (Fintech Hackathon); National Second Prize in the 14th Chinese Mathematics Competition (CMC)
 - **Scholarships:** 1st Place Scholarship 2024 (out of 174 students majoring in Economics); 2022 Excellent Scholarship
 - **Core Courses:** Artificial Intelligence and Python Programming (4.0), Calculus I (4.0), Calculus II (4.0), Linear Algebra (4.0), Probability Theory and Mathematical Statistics (4.0), Econometrics (4.0), Stata (4.0)
 - **Skills:** Linux, Python, R, SPSS, Stata, ArcGIS; Wind, Bloomberg; Photoshop, Animate
 - Exchange Program, University of California, Davis, CA, United States** 09/2023 - 12/2023
 - **GPA:** 4.00/4.00

WORK EXPERIENCE

- Teaching Assistant, Artificial Intelligence and Python Programming Course, Beijing, China** 09/2024 - 05/2025
- Link:** https://github.com/Quant-of-Renmin-University/QuantEcon2025_RUC
- Guided students to use different Python packages, such as Selenium, Scikit-learn and Keras, and assisted with grading
 - Maintained course GitHub projects and assisted in writing documentations and Latex codes for course slides
 - Quantitative Analyst Intern, Sinolink Securities, Shanghai, China** 01/2025 - 04/2025
 - Used Python to build quantitative investment models, such as risk parity and risk budgeting models; verified the validity of Dynamic-Nelson-Siegel (DNS) model in Chinese bond market
 - Built neural network, XGBoost models and fine-tuned hyperparameters to predict bond interest rates
 - Quantitative Analyst Intern, CITIC Securities Company Limited, Beijing, China** 07/2024 - 08/2024
 - Preprocessed CSI-300 data from 2010 to 2023, used fundamental and technical factors to build Random Forest and XGBoost model to predict stock price fluctuations, assessed factors by SHAP and mined deep factors using linear transformations to improve model performance

RESEARCH EXPERIENCE

- Research Assistant, Market Sentiment and IV Project, RUC Research Fund, Beijing, China** 03/2023 - 04/2024
- Processed 630,000 text data from traders' communities to construct investor sentiment indicators with the BERT model; predicted the implied volatility and set up the LSTM models to perform market sentiment analysis
 - Research Assistant, Enterprise and ESG Project, RUC School of Economics, Beijing, China** 01/2024 - 02/2024
 - Used Python and Excel to process over 200,000 enterprises and institutions data; determined the type of enterprises through text recognition and performed basic data analysis
 - Research Assistant, Real Estate and AI Project, RUC School of Economics, Beijing, China** 08/2024 - 12/2024
 - Analyzed and cleaned China's real estate big data in a Linux environment; designed a program to regularly update property information
 - Research Member, NCIETP—Gig Economy and Workers' Welfare, Beijing, China** 03/2023 - 04/2024
 - Used Credamo to distribute 300 questionnaires for takeout riders and collected their income, job satisfaction, etc.
 - Analyzed welfare and utility differences between two working modes; used Stata to conduct Logit regression to determine the main factors for workers to choose different working patterns

COMPETITION EXPERIENCE

- Best Modeling Award, RUC Quantitative Modeling Competition 2024, Beijing, China** 03/2024 - 06/2024
- Link:** <https://github.com/RUC-Tree/RUC-Fintech-Hackathon-2nd-Place-Code>
- Preprocessed raw housing data with Pandas and Re in Python; filled nulls with KNN; drew heat maps with Matplotlib to determine price trends; defined functions for feature engineering and generated deep features with OpenFE
 - Constructed linear models (OLS, Lasso, Ridge, Elastic Net), ensemble learning models (Random Forest, XGBoost), and Neural Network for training; used Optuna and GridSearch to search the optimal hyperparameters
 - Reduced MAE by applying Wide Neural Network (Inspiration from Cheng's 2016 paper); mixed multiple base models to reduce generalization errors; constructed counterfactual housing data to overcome data imbalance
 - National Second Prize, The 14th Chinese Mathematics Competition (CMC)** 11/2022
 - Examined advanced problems in partial differential equations, multivariable calculus, and infinite series.

PROJECT EXPERIENCE

- Best Prediction Project, Duke STA-521 "Nepal Earthquake" Project, Durham, NC** 12/2025
- Link:** <https://github.com/RUC-Tree/Duke-STA-521-Earthquake-Prediction>
- Built Geography AutoEncoder and compressed 12,568 categories to 16 "meaningful" vectors; utilized machine learning models (RF, XGBoost) to classify damage grades into 3 types and achieved the best performance in the class.
 - Fine-tuned the hyperparameters; analyzed feature importance and devised a "voting" model to improve performance.